Time Is the Baseline to Beat: Lead Time over a Timer as the Test for VLA Failure Monitors

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***Abstract*—On RoboCasa365, a failed vision-language-action (VLA) rollout is one that reaches its time limit. A runtime failure monitor therefore competes with a timer that alarms at a fixed fraction of that limit. Calibrated at a 5% false-positive rate (FPR), such a timer flags every one of 380 prospective failures, about 12 policy inferences before the timeout. Recall cannot separate a learned monitor from this timer; only lead time can. We test whether alarming when *either* a policy-internal score or the timer crosses its threshold warns earlier than time signals alone at a matched FPR, and calibrate this rule with a finite-sample FPR guarantee. On one VLA and 38 tasks, fusing the SAFE failure score with a single timer more than doubles the mean warning lead (0.090 to 0.207 of the time limit, exploratory), but most of that gain is task duration: a per-task timer, which alarms past the 95th percentile of each task's development success durations, nearly closes the gap without reading the policy. A preregistered test on a fresh cohort of 760 rollouts finds a small real gain over this per-task timer: 0.255 against 0.225 of the time limit (gain 0.030, 95% CI 0.007–0.093) at 0.061 FPR. Noise and within-task resampled scores fused the same way gain at most 0.009. The gain is above zero but below the margin of 0.05 we declared in advance, so the primary hypothesis is not supported. Stage 2 tests two learned monitors on a second VLA, and whether resetting at an alarm beats continuing within a fixed time budget.**

I. Introduction

Runtime monitors for VLA policies read the policy's hidden state and predict eventual failure [8], or score how unusual that state is relative to successful rollouts [27, 19]. They are compared with each other, but not with what the robot knows for free: how long the attempt has been running.

In RoboCasa365, as in most simulated manipulation benchmarks, an episode ends at first success or at a time limit, and in our test cohort every one of 380 failed rollouts ran to its limit. A timer at 90% of the limit therefore flags all of them at 4.7% FPR. Against it, a monitor can add only one thing: an earlier alarm at the same false-alarm rate, which leaves time to stop, reset or ask for help. We make this lead time the test, with three contributions.

1) **A testable hypothesis and endpoint.** We define the warning lead (the fraction of the time limit left at the alarm, zero if missed) and state four hypotheses with decision rules fixed in advance (Sec. IV).
2) **Calibrated OR-fusion with a guarantee.** We alarm when either a policy-internal score or the timer crosses its threshold, with a split-conformal calibration that bounds the FPR by α in finite samples (Proposition 1).
3) **Exploratory and confirmatory evidence.** SAFE fused with a single timer more than doubles its lead, but a per-task timer nearly matches it, and most failures are still recoverable at the limit (Sec. VI). On a fresh preregistered cohort the policy adds 0.030 of the limit over the per-task timer, below our 0.05 margin (Sec. VII).

The result is a small policy signal, not a useful one by our own standard: on timeout-defined benchmarks, policy internals add little to duration statistics, and monitors should be compared with a per-task timer and evaluated on failures that end early.

II. Related Work

Failure detection from policy internals. SAFE learns a failure score from a VLA's last-layer features and thresholds it with a functional conformal band [8]; we reuse its implementation. Follow-ups change the training data [18], labels [17, 14], features [28, 31] or policy [22]. None reports an elapsed-time detector. SAFE removes the duration shortcut by truncating rollouts to equal length and, on its real robot, by running every rollout to a fixed horizon [8]; a deployed monitor can do neither. Duration is a shortcut feature [7], but its causal form is legitimate online evidence.

Action-space and success-only detectors. STAC [1] and VLA-FAIL [20] compare overlapping action chunks, and ActProbe [9] feeds chunk consistency and magnitude to a task-conditioned LSTM, beating internal-feature baselines on RoboCasa. Our policy executes each chunk in full, so overlap signals are undefined. A task-conditioned recurrent detector can also learn how long each task takes, which our timers make explicit. FAIL-Detect [27] and FIPER [19] score inputs, outputs or embeddings against successful rollouts only. FIPER alarms when two conformal scores *both* exceed their thresholds, which suppresses false alarms; in its own pipeline an elapsed-time method matches its accuracy, but FIPER warns earlier at a matched false-alarm rate (Sec. VI). We alarm when *either* crosses, because the policy score and the timer catch failures at different times, and calibrate the pair jointly.

World-model verifiers and VLM monitors. Check-VLA [12] is the closest work on our benchmark. It verifies each action chunk with an action-conditioned world model on RoboCasa365, at a conformal 5% rate of unnecessary interventions, and repairs it with the same VLA. Foresight [24] uses world-model latents. Neither compares against elapsed time; either could be fused with a timer. LLM and VLM monitors [1, 13, 21, 30, 6] are slower and complementary.

Time, calibration and intervention. VLA confidence drifts over a task [32]. Our time-risk baseline is a per-task survival-style estimate [10, 5], and lead time at a fixed false-alarm budget is the accuracy–earliness trade-off of early time-series classification [26]. Split-conformal thresholds give finite-sample false-positive control under exchangeability [3]; we extend the construction to a two-signal OR rule (Sec. V-C). RESCUE [2] finds that good detection scores do not guarantee lower intervention cost, and CheckVLA and FLARE [29] couple detection to repair; an earlier alarm matters only if acting on it raises success.

III. Problem Setting

Time. The simulator runs *control steps* at RoboCasa's default 20 Hz. The policy executes each 16-step action chunk in full, so *policy inferences* $t = 1, 2, \ldots$ (0.8 s each) are our clock, and a monitor decides only at an inference. Task k's limit of T_k control steps is $H_k = \lceil T_k/16 \rceil$ inferences (29 to 272, or 22.5 to 217.5 s). Every fraction of time (t/H_k, c, ℓ) is a fraction of H_k.

Monitoring. A monitor watches a rollout of a known task k and decides at each inference t whether to alarm, using only the policy features so far, t, the task and H_k. The first crossing is the alarm T (in inferences). FPR is the fraction of successful rollouts with any alarm. A rollout of length L inferences ends at success or at H_k; for a failure ($L = H_k$) the *warning lead* is $\ell = (L - T)/H_k$, or 0 if missed, and an alarm at the final inference counts with $\ell = 0$. The mean lead $\bar{\ell}$ combines recall and earliness. Detectors are compared at operating points fixed on a calibration cohort under FPR $\leq \alpha = 0.05$. Confidence intervals come from a paired bootstrap that resamples tasks, then rollouts within each task (2,000 replicates).

Scope. Whether elapsed time is a strong baseline depends on the termination rule. Where successes stop at completion (RoboCasa, FIPER's tasks, deployment), it is. Where every rollout runs to one fixed horizon, as in SAFE's real-robot set (in its release, 98% of failures and ≥ 30 successes per task stop at one step) or SimplerEnv, it is uninformative, and early-warning claims need failure-time annotations, which SAFE did not release.

IV. Hypotheses

We test four hypotheses. H1 and H2 were evaluated on a fresh confirmatory cohort with the decision rules below, registered as an amendment before any timer or lead quantity was computed on that cohort (Sec. VII); H3 and H4 are the Stage 2 extensions (Sec. VIII).

H1 (lead time, primary). At thresholds calibrated for FPR ≤ 0.05, OR-fusing a policy-internal score with the per-task timer (Sec. V), the strongest time-only monitor we know, increases the mean warning lead over that timer alone. *Supported* if the paired bootstrap 95% CI of $\Delta\bar{\ell}$ lies above zero, the point gain is at least 0.05 of the time limit, and the fused test FPR is at most 0.075 with a Wilson 95% upper bound at most 0.10. As a secondary check, the fused detector must also beat a timer whose fraction is tuned *on the test cohort* to the fused detector's test FPR, which favours the timer.

H2 (the gain needs policy evidence). Fusions that contain no policy-internal evidence (Gaussian noise, or SAFE scores resampled within task, each fused with the per-task timer) gain less than 0.05 of the time limit over the per-task timer.

H3 (second policy). On RLDX-1's 20 tasks, each of two co-primaries, SAFE + P and Hide-and-Seek + P, increases the mean lead over P: its 97.5% decision CI (Bonferroni over the two) lies above zero under the decision rule of E2. There is no 0.05 margin, and inference is conditional on these 20 tasks; generalization to other tasks is a secondary with no power claim.

H4 (utility). Within a budget of $1.25\,H_k$, resetting the scene and retrying when P alarms yields a higher task success rate than letting the attempt continue to the same budget.

Only H1 and H2 are confirmatory; every other interval in this paper is descriptive. We expected H1 to fail (opened-data gain 0.007); it did, with a gain of 0.030 whose CI excludes zero (Sec. VII).

V. Method

A. Signals

Policy score. At each inference we take the final-layer action-token hidden state of the policy's action head (1,024 dimensions). The SAFE MLP [8] maps it to a failure score $u_t^{(j)}$ for each of three training seeds j. With per-task location $m_k^{(j)}$ and scale $\sigma_k^{(j)}$ estimated on development rollouts, the policy score is

$$s_t = \frac{1}{3}\sum_{j=1}^{3} \frac{\max_{t' \leq t} u_{t'}^{(j)} - m_k^{(j)}}{\sigma_k^{(j)}}. \tag{1}$$

The running maximum makes s_t monotone and does not change when a threshold is first crossed. Success-only alternatives (Mahalanobis [11], kNN [23], PCA-KMeans and RND [4]) are fitted on successful development rollouts and aggregated the same way.

Timers. The global timer alarms once $t \geq c\,H_k$ for a fraction $c \in (0, 1]$ shared by all tasks; it uses only the task's time limit. The per-task timer uses $c_k = \min(1, \rho_k + \delta)$, where ρ_k is the 95th percentile of task k's development success durations as fractions of H_k and the shared margin δ is set by Proposition 1 on calibration successes. It needs successful rollouts of each task, but no failures and no policy features. We fixed the 95th percentile after auditing the opened cohort, so we report 0.75, 0.90 and 1.0 (the longest success) as sensitivity.

Time risk (baseline). The survival-style alternative estimates $r_t = \Pr(\text{failure} \mid k, \text{still running at } t)$ from development rollouts: with n_t of them still running at t and f_t of those failing, it fits an increasing isotonic regression in t/H_k to $(f_t + 0.5)/(n_t + 1)$, weighted by n_t. As every failure runs to H_k, r_t is a task-specific timer, but its smoothing keeps it below a high threshold in tasks with few development failures.

B. OR-fusion

The fused monitor alarms at

$$T = \min\{\, t : s_t > \tau_s \text{ or } t \geq c\, H_k \,\}. \tag{2}$$

Neither signal is restricted in time: the policy score may alarm early, and the timer guarantees an alarm on every failure, at $c\,H_k$ at the latest. The lead gain over the timer comes only from failures on which the policy score crosses first.

C. Calibration with a finite-sample FPR guarantee

A successful rollout i of length L_i (in inferences) raises a false alarm under (τ_s, c) if and only if $M_i = \max_{t \leq L_i} s_t > \tau_s$ or $\rho_i = L_i/H_{k_i} \geq c$, so its false alarm depends only on the pair (M_i, ρ_i). Fix in advance a path $\gamma(\lambda) = (\tau_s(\lambda), c(\lambda))$, $\lambda \in [0, 1]$, with both coordinates continuous and nondecreasing, so that a larger λ is more conservative and $\gamma(1)$ never alarms on a success. For each calibration success define

$$V_i = \min\{\lambda : M_i \leq \tau_s(\lambda) \text{ and } \rho_i < c(\lambda)\}, \tag{3}$$

the least conservative point on the path at which rollout i stays silent. Because the path is monotone, rollout i is silent at λ exactly when $\lambda \geq V_i$.[1]

Proposition 1: Let n calibration successes and a new successful rollout be exchangeable, and let the path γ not depend on the calibration successes. Set $\hat{\lambda} = V_{(\lceil (n+1)(1-\alpha) \rceil)}$, the $\lceil (n+1)(1-\alpha) \rceil$-th smallest V_i (or $\hat{\lambda} = 1$ if that rank exceeds n). Then the OR-fused monitor at $\gamma(\hat{\lambda})$ raises a false alarm on the new success with probability at most α.

Proof: The new rollout alarms at $\hat{\lambda}$ if and only if $V_{n+1} > \hat{\lambda} = V_{(r)}$ with $r = \lceil (n+1)(1-\alpha) \rceil$. By exchangeability, the rank of V_{n+1} among $V_1, \ldots, V_{n+1}$ is uniform on $\{1, \ldots, n+1\}$ (ties broken at random), so $\Pr(V_{n+1} > V_{(r)}) \leq 1 - r/(n+1) \leq \alpha$; this is the split-conformal quantile lemma [3] applied to the scalar V. ∎

The path may use any data except calibration successes, such as development rollouts or calibration *failures*. The guarantee is marginal over tasks; a per-task version needs 19 calibration successes per task.

The preliminary results use the original protocol's joint search: among all pairs of calibration maxima with calibration FPR $\leq \alpha$, keep the one with the largest mean lead (highest recall for time-risk fusions). This best-of-many selection on 114 successes has no guarantee and is optimistic (test FPR 0.058–0.068 against 0.044 on calibration). The per-task timer rows, and the confirmatory test, use Proposition 1 with the path $\delta = \lambda$, $\tau_s = u + s\lambda$ and (u, s) chosen on calibration failures.

VI. Preliminary Evidence

Setup. We use the released Xiaomi-Robotics-1 RoboCasa365 checkpoint [25], which executes each 16-step action chunk in full, on the 38 official RoboCasa365 tasks [15, 16] with published success between 5 and 45 of 50 and enough failures (10 atomic, 28 composite; failures last 29–272 inferences). Disjoint cohorts serve one purpose each: development (1,824 rollouts) fits SAFE and the time risk; calibration (3 successes and 3 failures per task; 454 rollouts including those past the quota) sets thresholds; the prospective test (10 per class per task) is evaluated once; a confirmatory cohort of the same size is used only in Sec. VII. SAFE, its normalization and the time risk were frozen before the test was collected; the timers and fused detectors were added later, with thresholds from calibration only, so the results below are exploratory.

[1]The strict $\rho_i < c(\lambda)$ makes the set right-open; taking c on a finite grid, as the timer's resolution of one inference does anyway, makes the minimum attained.

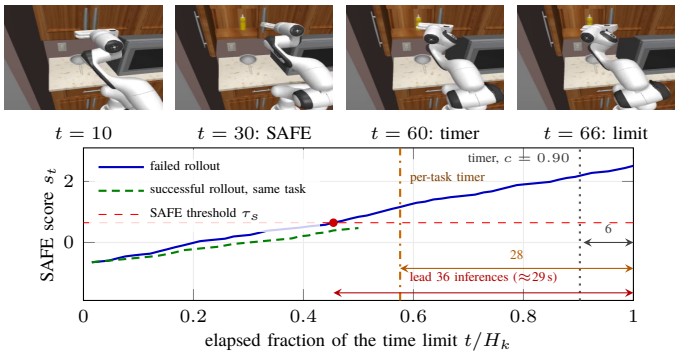


Fig. 1. One failed *OpenCabinet* rollout from the prospective test (chosen to illustrate; the median SAFE-first lead is smaller). The SAFE score crosses its calibrated threshold at $t = 30$ of $H_k = 66$ inferences, 36 inferences before the time limit; the timer alarms at $t = 60$, and the per-task timer, which never reads the policy, at $t = 38$. Dashed: the longest of the 9 (out of 10) successful test rollouts of this task that stay below the threshold; it ends at $t = 33$.

TABLE I
PROSPECTIVE TEST (380 SUCCESSES, 380 FAILURES), THRESHOLDS FROM CALIBRATION AT FPR ≤ 0.05. $\bar{\ell}$: MEAN LEAD (MISS = 0); MED: MEDIAN LEAD (INFERENCES) OF DETECTED FAILURES; @.75: FAILURES ALARMED BY $0.75\,H_k$. *MEAN OF 20 DRAWS ($\bar{\ell}$ 0.088–0.134). †MARGIN BY PROPOSITION 1.

Detector	Recall	FPR	$\bar{\ell}$	Med	@.75
Timer ($c = 0.903$)	1.000	0.047	0.090	11.5	0.000
Time risk	0.632	0.047	0.120	16.5	0.211
Time risk + timer	1.000	0.050	0.125	11.0	0.105
Noise + timer*	1.000	0.066	0.101	–	–
SAFE only	0.566	0.039	0.190	23.0	0.421
SAFE + time risk	0.955	0.058	0.184	12.0	0.311
Per-task timer†	1.000	0.045	0.195	13.5	0.342
SAFE + per-task timer†	1.000	0.042	0.202	14.0	0.389
SAFE + timer	1.000	0.058	0.207	10.0	0.421

The timer is the baseline to beat. The timer ($c = 0.903$) alarms on all 380 test failures at 0.047 FPR with a fixed lead of 0.090 (Table I). The time-risk curve misses 140 failures, all in 14 tasks whose smoothed curve never reaches its threshold, so recall gains measured against it (to 0.90–0.96 for SAFE and four success-only scores) vanish here.

SAFE adds lead over a single timer. SAFE + timer raises the mean lead to 0.207 (gain 0.117, CI 0.060–0.178) and flags 42% of failures by $0.75\,H_k$ (Figs. 1, 2) at 0.058 FPR; noise + timer reaches 0.101. Among rollouts still running at 10, 25, 50 and 75% of the limit, SAFE's within-task ROC-AUC is 0.55–0.60, against 0.41–0.51 for noise.

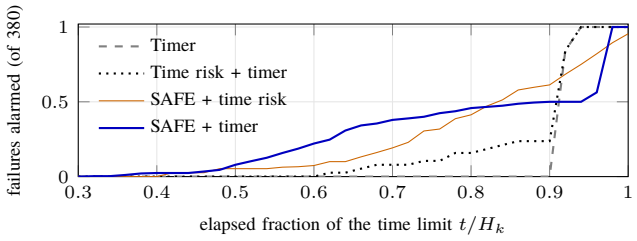


Fig. 2. Fraction of the 380 test failures alarmed by each elapsed fraction of the time limit. SAFE + timer catches half of the failures early and the rest at its timer fallback ($c = 0.96$).

TABLE II

CONFIRMATORY TEST (380 SUCCESSES, 380 FAILURES). P = PER-TASK TIMER ($q = 0.95$). H1 INTERVAL: REGISTERED DECISION BOOTSTRAP; OTHERS: TEST ONLY. *MEAN OF 20 DRAWS [MAXIMUM]. †FROZEN ORIGINAL THRESHOLDS.

Detector	Recall	FPR	$\bar{\ell}$	Gain vs P
Global timer	1.000	0.024	0.123	-0.102 (-0.147, -0.059)
SAFE + time risk†	0.955	0.039	0.190	-0.035 (-0.066, -0.007)
SAFE + global timer	1.000	0.047	0.215	-0.011 (-0.068, 0.045)
P	1.000	0.055	0.225	–
Noise + P*	–	–	–	0.002 [0.009]
Resampled SAFE + P*	–	–	–	0.002 [0.007]
SAFE + P (H1)	1.000	0.061	**0.255**	**0.030** (0.007, 0.093)

A per-task timer removes the gain (H1, exploratory). The per-task timer reaches $\bar{\ell} = 0.195$ at 0.045 FPR without reading the policy. SAFE + per-task timer reaches 0.202, a gain of 0.007 (CI -0.003 to 0.019), far below the 0.05 margin of H1.

No early warning, and mostly no early failure (exploratory). No detector flags over 8% of failures by half the limit. Is timeout-defined failure then trivial by construction? One labeler, blind to alarms, marked the point of no return (after which the policy could not finish in time without help) on 48 confirmatory-cohort failures (2 per atomic, 1 per composite task): 38 (79%) were still recoverable at the limit (atomic 17/20, composite 21/28), most often slow or nearly successful. A second labeler agreed on 11 of 12 (Cohen's $\kappa = 0.80$). Failure here mostly means not finishing in time, which is why elapsed time is so informative; an early "failure" alarm mostly predicts slowness. On the 10 decided failures, P alarmed after the point of no return on all, SAFE + P at or before it on 2 of 3 atomic and 0 of 7 composite ones (an illustration).

A second policy, offline (RLDX-1, exploratory). On existing RLDX-1 data (20 tasks; 740 development rollouts, 60 held-out calibration successes, 100/100 test), all 438 failures ran to the limit. P reaches $\bar{\ell} = 0.271$ at 0.03 test FPR, the global timer 0.032. SAFE (LSTM head) + P gains 0.034 over P (CI 0.008–0.067) at the *same* FPR, so none of it is extra FPR budget; the MLP head adds nothing. The gain is atomic (0.049, CI 0.013–0.097); on composite tasks it is 0, as SAFE never alarms before P. Caveats: the fused path was chosen on in-sample development failures, the FPR margin rests on 60 successes, and noise + P matched the gain in 2 of 20 draws, both at 0.08 FPR.

The test on a published benchmark (exploratory). In FIPER's released rollouts [19] (five tasks, two on a real robot), length alone separates failures with ROC-AUC 0.92–1.00 in four tasks (0.62 in the fifth). We re-ran FIPER's own code, which reproduces its published results within 0.05, adding one method whose score is elapsed time. Under FIPER's thresholds and selection rule, the timer has the best balanced accuracy (0.81 against 0.78) and a test FPR of 0.09, against 0.35 for FIPER. At FIPER's FPR, however, FIPER warns 0.146 of the episode earlier than the timer (CI 0.104–0.188; post hoc), on four of five tasks. The test thus credits FIPER, but finds SAFE's gain on XR-1 small (Sec. VII): it discriminates both ways.

VII. CONFIRMATORY TEST OF H1 AND H2

Protocol. A fresh test cohort (new seeds, 10 per class per task) was collected with SAFE frozen and preregistered for recall against the time risk. Before any timer or lead was computed on it, but with its recall and FPR known, an amendment made the per-task timer P ($q = 0.95$) the comparator and $\Delta\bar{\ell}$ the endpoint, with the rules of Sec. IV. A new calibration cohort of the same design (380 successes) sets the thresholds of P and SAFE + P by Proposition 1. The decision interval resamples both cohorts and refits thresholds in each of 2,000 replicates. Every test failure again ran to H_k.

H1 is not supported, but the gain is not zero. SAFE + P leads P by 0.030 of the time limit (Table II). Its decision CI lies above zero and its test FPR (0.061, Wilson 95% upper bound 0.089) meets the FPR rules, but the point gain is below the 0.05 margin, so H1 fails. The CI (upper end 0.093) does not show the gain is negligible, and the opened-cohort estimate (0.007) understated it.

H2 is supported. Noise and within-task resampled SAFE scores, fused with P and calibrated the same way, gain at most 0.009 over P in 20 draws each. What SAFE adds is policy evidence, not an artifact of fusing any score with a timer.

Task duration carries most of the lead. P alone leads the global timer by 0.102, about what SAFE + global timer reaches, and the original SAFE + time-risk design, which beat the time risk on recall here, gives 0.035 *less* lead than P. Part of the H1 gain is FPR budget: a per-task timer tuned on the test cohort to SAFE + P's FPR leaves a gain of only 0.015 (CI -0.007 to 0.041), and a global timer tuned the same way 0.024 (CI -0.022 to 0.074), so the secondary check of H1 also fails. The gain is 0.026–0.030 for $q = 0.75$–0.95 and 0.074 at $q = 1$, where P itself is weaker (0.164).

Atomic tasks carry the gain (descriptive). On the 10 atomic tasks SAFE + P gains 0.081 (0.372 against 0.291); on the 28 composite tasks, 0.012. Exploratory, on the same cohorts: the four frozen success-only scores fused with P gain 0.020–0.034 (SAFE 0.030), so on lead, as on recall, the choice of score matters little; and all five gain more on atomic (0.045–0.081) than on composite tasks (0.008–0.027). The split is a property of atomic tasks under the per-task timer, not of

SAFE, and RLDX-1 shows it too. On composite tasks P likely lacks progress: on five, an oracle reading subtask predicates gained 0.071 over P (task-level CI -0.021 to 0.129). FPR is not controlled per stratum.

VIII. Stage 2 Plan

E2: Second policy (H3). RLDX-1 on its 20 tasks, fresh calibration and test cohorts of 20 successes and 20 failures per task (about 2,500 rollouts). Calibration failures are held out, so the fused path is not chosen in-sample. Frozen before collection, from the existing data (Sec. VI): SAFE's LSTM head (its MLP head falls back to the timer offline); Hide-and-Seek [17], a supervised early-separation probe on the hidden states (three seeds); H_k, the longest development failure of task k; and ρ_k, the 95th percentile of its development success durations over H_k.[PENDING: checkpoint hashes] Hide-and-Seek + P gained 0.071 over P in a preregistered test on the E1 cohorts and 0.083 offline on RLDX-1. *Decision*, per co-primary: the lead gain over P has a 97.5% decision CI above zero (rollouts resampled within task), and so does its gain over P tuned on test to the fused detector's own FPR, so that false-alarm budget cannot pass the test alone.[TODO: matched-FPR condition proposed; author to confirm] Before collection we simulate this rule's type I error under noise + P and within-task resampled scores + P, and its power; $n = 20$ unless that requires more.[PENDING: type I error, power, final n] Without the matched-FPR condition, power at $n = 20$ is at least 0.98 for each. Secondary, no power claim: the CI with tasks resampled. Descriptive: the 14 atomic tasks. Sizing is optimistic (5 held-out failures per task).

FPR rule.[PENDING: author to confirm the FPR rule] Calibration targets $\alpha = 0.05$; false alarms must not reject "true FPR ≤ 0.075" in a one-sided exact binomial test at level 0.05 (at most 39 of 400 in E2); H1's fixed limit rejects good detectors in small cohorts.

E3: Reset and retry (H4). On XR-1's 38 tasks[PENDING: policy: XR-1 proposed, author to confirm], each episode has a budget of $1.25\,H_k$. *Primary*: when P alarms, the scene is reset to its initial state for a new attempt, against continuing the same attempt to $1.25\,H_k$; as most failures are still recoverable at the limit, continuation is the decisive comparator.[TODO: primary contrast proposed; author to confirm] Secondary: P-triggered reset against no monitor (reset only at H_k). Each environment seed runs under every arm, a retry reuses the scene with a new policy sampling seed, and success is compared by McNemar's test on discordant pairs. SAFE + P is descriptive (at least about 1,450 episodes per arm to separate it from P). The secondary needs at least about 650 episodes per arm if a retry succeeds at the task's success rate, which is optimistic for the same scene; a pilot of about 100 re-attempts of failed XR-1 scenes estimates it and sets the final size.[PENDING: pilot result; final size]

E4, E5 (exploratory, no decision rules). E4: action-only monitors, richer time baselines, unseen tasks. E5: lead from the point of no return, fused detector vs P, on about 250–500 labeled failures (50–100 decided), with third-labeler adjudication, reported only if a pilot of ≥ 20 doubly labeled decided failures reaches mean $|\Delta t_u|/H_k \leq 0.10$.[TODO: E5 exploratory proposed; registering it is the author's call]

IX. Limitations

These are simulated shadow evaluations: one policy prospectively, RLDX-1 only offline on a small test set. Timers and fused detectors were defined after the exploratory cohort was opened, and the confirmatory endpoint after its recall and FPR were known. Task and limit are assumed known; cohorts are quota-balanced. Point-of-no-return labels are one labeler's judgments (12 checked by a second); 10 failures were decided. The FIPER lead comparison is post hoc, with 10 + 10 real-robot test rollouts per task (PushChair: a lower bound).

X. Conclusion

When failure means timeout, and mostly slowness, a learned monitor must warn earlier than a matched-FPR timer, as FIPER does on its benchmark. On XR-1, SAFE added 0.030 of the time limit to a per-task timer, below our declared margin; Stage 2 tests two monitors on a second policy, and closed-loop retry.

References

[1] Christopher Agia, Rohan Sinha, Jingyun Yang, Zi-ang Cao, Rika Antonova, Marco Pavone, and Jeannette Bohg. Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress. In *Conference on Robot Learning (CoRL)*, volume 270 of *PMLR*, 2024.

[2] David Ahmedt-Aristizabal, Abdelwahed Khamis, Mohammad Ali Armin, and Lars Petersson. RESCUE: Is this alarm worth acting on? Counterfactual evaluation of VLA runtime monitors, 2026. arXiv:2610.13516.

[3] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.

[4] Yuri Burda, Harrison Edwards, Amos Storkey, and Oleg Klimov. Exploration by random network distillation. In *International Conference on Learning Representations (ICLR)*, 2019.

[5] David R. Cox. Regression models and life-tables. *Journal of the Royal Statistical Society: Series B*, 34(2):187–220, 1972.

[6] Jiafei Duan, Wilbert Pumacay, Nishanth Kumar, Yi Ru Wang, Shulin Tian, Wentao Yuan, Ranjay Krishna, Dieter Fox, Ajay Mandlekar, and Yijie Guo. AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation. In *International Conference on Learning Representations (ICLR)*, 2025.

[7] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2 (11):665–673, 2020.

[8] Qiao Gu, Yuanliang Ju, Shengxiang Sun, Igor Gilitschenski, Haruki Nishimura, Masha Itkina, and Florian Shkurti. SAFE: Multitask failure detection for vision-language-action models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[9] Bingjia Huang, Xiangyu Li, Xiang Wang, Liang Mi, Zixu Hao, Weijun Wang, Hao Wu, Kun Li, Yunxin Liu, and Ting Cao. ActProbe: Action-space probe for early failure detection of generative robot policies, 2026. arXiv:2606.08508.

[10] Edward L. Kaplan and Paul Meier. Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association*, 53(282):457–481, 1958.

[11] Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[12] Yushan Liu, Peibo Sun, Xintao Chao, Zhenyang Yang, Yifan Xie, Lingfeng Zhang, Shoujie Li, Chenyu Tang, Fang Chen, Xiao-Ping Zhang, and Wenbo Ding. CheckVLA: Execution-time verification with action-conditioned world model for long-horizon mobile manipulation, 2026. arXiv:2607.26789.

[13] Zeyi Liu, Arpit Bahety, and Shuran Song. REFLECT: Summarizing robot experiences for failure explanation and correction. In *Conference on Robot Learning (CoRL)*, 2023.

[14] Jie Ma, Zongxi Liu, and Yi Zhu. FailureSpot: Label-efficient timestamp-level failure detection for vision-language-action models, 2026. arXiv:2609.04277.

[15] Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. RoboCasa: Large-scale simulation of everyday tasks for generalist robots. In *Robotics: Science and Systems (RSS)*, 2024.

[16] Soroush Nasiriany, Sepehr Nasiriany, Abhiram Maddukuri, and Yuke Zhu. RoboCasa365: A large-scale simulation framework for training and benchmarking generalist robots. In *International Conference on Learning Representations (ICLR)*, 2026.

[17] Seongheon Park, Wendi Li, Changdae Oh, Min-Hsuan Yeh, Zsolt Kira, Michael Hagenow, and Sharon Li. Hide-and-seek in trajectories: Discovering failure signals for VLA runtime monitoring. In *ICML Workshop on Failure Modes in Agentic AI*, 2026. arXiv:2605.30834.

[18] Harshitha Rajaprakash, Aditeya Prajapati, Rong Xue, Abrar Anwar, and Jesse Thomason. SAFECAST: Robust failure detection for VLA policies with contrast-set training and calibration, 2026. arXiv:2608.04246.

[19] Ralf Römer, Adrian Kobras, Luca Worbis, and Angela P. Schoellig. Failure prediction at runtime for generative robot policies. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[20] Florian Seligmann, Emiliyan Gospodinov, Enes Ulas Dincer, and Gerhard Neumann. VLA-FAIL: Efficient task failure detection for finetuned vision-language-action models, 2026. arXiv:2606.21386.

[21] Rohan Sinha, Amine Elhafsi, Christopher Agia, Matthew Foutter, Edward Schmerling, and Marco Pavone. Real-time anomaly detection and reactive planning with large language models. In *Robotics: Science and Systems (RSS)*, 2024. doi: 10.15607/RSS.2024.XX.114.

[22] Michael Soddy, Karthikeyan Ganesan, Roman Stemp, Ved Deo, and Nicholas Babey. Predictable before action, resistant to correction: Failure signals in vision-language-action models. In *NeurIPS Workshop on From Pretrained Representations to Acting Agents (PTA)*, 2026.

[23] Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In *International Conference on Machine Learning (ICML)*, 2022.

[24] Boyang Wang, Haoran Zhang, Mengdi Wang, Odest Chadwicke Jenkins, Xuhui Kang, Yen-Ling Kuo, Yifu Lu, and Zezhou Cheng. Foresight: Failure detection for long-horizon robotic manipulation with action-conditioned world model latents, 2026. arXiv:2606.23085.

[25] Xiaomi Robotics Team. Xiaomi-Robotics-1: Scaling vision-language-action models with over 100K hours of real-world trajectories, 2026. arXiv:2607.15330.

[26] Zhengzheng Xing, Jian Pei, and Philip S. Yu. Early classification on time series. *Knowledge and Information Systems*, 31(1):105–127, 2012. doi: 10.1007/s10115-011-0400-x.

[27] Chen Xu, Tony Khuong Nguyen, Emma Dixon, Christopher Rodriguez, Patrick Miller, Robert Lee, Paarth Shah, Rares Ambrus, Haruki Nishimura, and Masha Itkina. Can we detect failures without failure data? Uncertainty-aware runtime failure detection for imitation learning policies. In *Robotics: Science and Systems (RSS)*, 2025.

[28] Jinghan Yang, Yunchao Zhang, Wang Yuan, Haolun Wan, Jiaming Zhang, Zhengyang Hu, and Yanchao Yang. Tri-Info: Generalizable, interpretable failure prediction for VLA models via information theory, 2026. arXiv:2606.19998.

[29] Ganlong Zhao, Zijia Tang, Xingping Chen, Zhanghui Kuang, Ye Tian, and Guanbin Li. FLARE: A failure-aware framework for autonomous correction and recovery in visual-language robotic manipulation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22391–22401, 2026.

[30] Enshen Zhou, Qi Su, Cheng Chi, Zhizheng Zhang, Zhongyuan Wang, Tiejun Huang, Lu Sheng, and He Wang. Code-as-monitor: Constraint-aware visual programming for reactive and proactive robotic failure detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6919–6929, 2025.

[31] Kaiwen Zhu, Dongfang Liu, and Liangkai Liu. VLA-Scope: Shift-aware failure prediction for vision-language-

action models, 2026. arXiv:2609.21246.

[32] Tom Zollo and Richard Zemel. Confidence calibration in vision-language-action models. In *ICML Workshop on Statistical Frameworks for Uncertainty in Agentic Systems*, 2026. arXiv:2507.17383.